

Project Title: Robustness Enhancement in Image Classification via Data Augmentation

Subject: DASC 5304 Machine Learning

Project Proposal by: Jagriti Koirala (UTA ID: 1002340217), Pritesh Das (UTA ID: 1002333230)

Dataset: CIFAR-10, CIFAR-100, and SVHN image classification datasets

Project Idea: For this project, we aspire to study how different data augmentation approaches affect the accuracy and robustness of convolutional neural networks (CNNs) on CIFAR-10, CIFAR-100, and SVHN. We will first train a baseline CNN on each dataset with simple augmentations (random crop, horizontal flip), then add stronger techniques such as color jitter, Cutout, Mixup, and CutMix, comparing their impact across datasets using a consistent model architecture and training protocol.

GAN-based data augmentation will be implemented by training a generative adversarial network on CIFAR-10 to synthesize additional class-conditional images, allowing us to compare models using only classical and mix-based augmentations to those that also use GAN generated samples on CIFAR-10, while CIFAR-100 and SVHN will serve to study how augmentation strategies transfer to datasets with higher class complexity and a different domain without GAN generated data. Our main questions are which augmentation combinations best balance accuracy and robustness across these datasets, and whether GAN based augmentation on CIFAR-10 provides tangible robustness gains over classical augmentation alone.

Software & Tools: Python, PyTorch, torchvision, NumPy, pandas, Matplotlib, Seaborn, Google Colab, Jupyter Notebook

Computing Resources: Training CNNs on CIFAR-10, CIFAR-100, and SVHN is feasible on a single modern GPU, using either university devices or cloud platforms such as Google Colab. The datasets allow CNN/ResNet experiments with classical and mix-based augmentations to run comfortably on GPUs with 8–16 GB of memory. GAN training will be restricted to CIFAR-10 to keep the computational load manageable, and this remains viable due to the small image size and dataset scale. Standard laptop storage is sufficient for storing all three datasets, model checkpoints, and experimental logs.

Milestone: By the halfway mark we plan to have a baseline CNN/ResNet trained on CIFAR-10, CIFAR-100, and SVHN with standard augmentations and reported clean test accuracy for each dataset, to have added and evaluated at least one stronger augmentation method (such as Mixup or CutMix) on all three datasets with initial comparisons to the baseline models, to have obtained initial CIFAR-10-C (corrupted version of CIFAR-10) robustness results for the CIFAR-10 baseline and stronger-augmentation models, and to have trained an initial GAN on CIFAR-10 with initial synthetic samples integrated into at least one CIFAR-10 classifier run, providing the required first set of experimental results.

References and Papers to Read:

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4. T. Karras et al., "Training generative adversarial networks with limited data," NeurIPS, 2020.
5. S. K. Y. Hong, T. Salimans, H. Zhang, and D. P. Kingma, "Differentiable augmentation for data-efficient GAN training," NeurIPS, 2020.
6. A. F. M. Agustsson et al., "Masked generative adversarial networks are data-efficient learners," arXiv, 2020.
7. S. Khan, A. Sohail, and A. S. Qureshi, "A comprehensive review on deep convolutional neural networks," IEEE Transactions on Neural Networks and Learning Systems, 2022.
8. J. Yang, X. Chen, and L. Zhang, "GAN-based data augmentation for image classification: A survey and empirical study," Engineering Applications of Artificial Intelligence, 2023.
9. Y. Zhang, H. Wang, and J. Li, "A survey of mix-based data augmentation: Taxonomy, methods, applications, and explainability," ACM Computing Surveys, 2024.
10. S. El-Fakharany, A. M. Sarhan, and A. M. El-Gayar, "Image classification using deep learning: A comparative study," IEEE, 2021.